Packet Sniffer & Analyzer using Python

Platform: Windows
Language: Python
Library: Scapy

1. Project Title

Packet Sniffer & Analyzer using Python

2. Objective

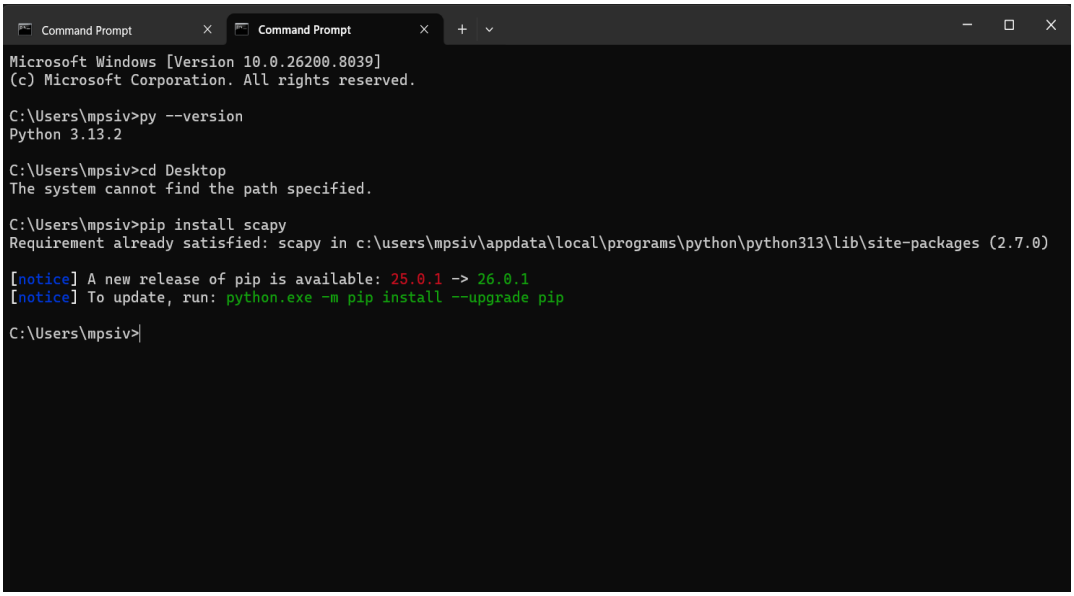
The objective of this project is to capture and analyze network packets in real-time using Python and Scapy.

3. Software Requirements

- Windows OS
- Python 3.x
- Command Prompt
- Scapy library

4. Step 1 – Check Python Installation

Command: py --version



5. Step 2 – Install Scapy

Command: pip install scapy

```
Command Prompt    Command Prompt
The system cannot find the path specified.

C:\Users\mpsiv>pip install scapy
Collecting scapy
  Downloading scapy-2.7.0-py3-none-any.whl.metadata (5.8 kB)
Downloading scapy-2.7.0-py3-none-any.whl (2.6 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2.6/2.6 MB 19.7 MB/s eta 0:00:00
Installing collected packages: scapy
Successfully installed scapy-2.7.0

[notice] A new release of pip is available: 25.0.1 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip

C:\Users\mpsiv>notepad sniffer.py

C:\Users\mpsiv>py sniffer.py
Sniffing packets... Press Ctrl+C to stop
Ether / IPv6 / TCP 2404:6800:4007:804::200e:https > 2406:7400:ff03:fb2d:9c21:41ef:8e77:dc73:56424 PA / Raw
Ether / IPv6 / TCP 2404:6800:4007:804::200e:https > 2406:7400:ff03:fb2d:9c21:41ef:8e77:dc73:56424 FA
Ether / IPv6 / TCP 2406:7400:ff03:fb2d:9c21:41ef:8e77:dc73:56424 > 2404:6800:4007:804::200e:https A
Ether / IPv6 / UDP 2406:7400:ff03:fb2d:9c21:41ef:8e77:dc73:53762 > 2404:6800:4007:835::200a:https / Raw
Ether / IPv6 / UDP 2404:6800:4007:835::200a:https > 2406:7400:ff03:fb2d:9c21:41ef:8e77:dc73:53762 / Raw
Ether / IPv6 / TCP 2a06:98c1:3107::6812:2715:https > 2406:7400:ff03:fb2d:9c21:41ef:8e77:dc73:53015 PA / Raw
Ether / IPv6 / TCP 2406:7400:ff03:fb2d:9c21:41ef:8e77:dc73:53015 > 2a06:98c1:3107::6812:2715:https PA / Raw
Ether / IPv6 / TCP 2a06:98c1:3107::6812:2715:https > 2406:7400:ff03:fb2d:9c21:41ef:8e77:dc73:53015 A
Ether / IPv6 / ICMPv6ND_RA / ICMPv6NDOptPrefixInfo / ICMPv6NDOptRouteInfo / ICMPv6NDOptMTU / ICMPv6 Neighbor Discovery O
ption - Source Link-Layer Address 8c:86:dd:5a:cd:78
Ether / IPv6 / TCP 2406:7400:ff03:fb2d:9c21:41ef:8e77:dc73:64798 > 2404:6800:4007:83a::200e:https A / Raw

C:\Users\mpsiv>
```

6. Step 3 – Create Python File

Command: notepad sniffer.py

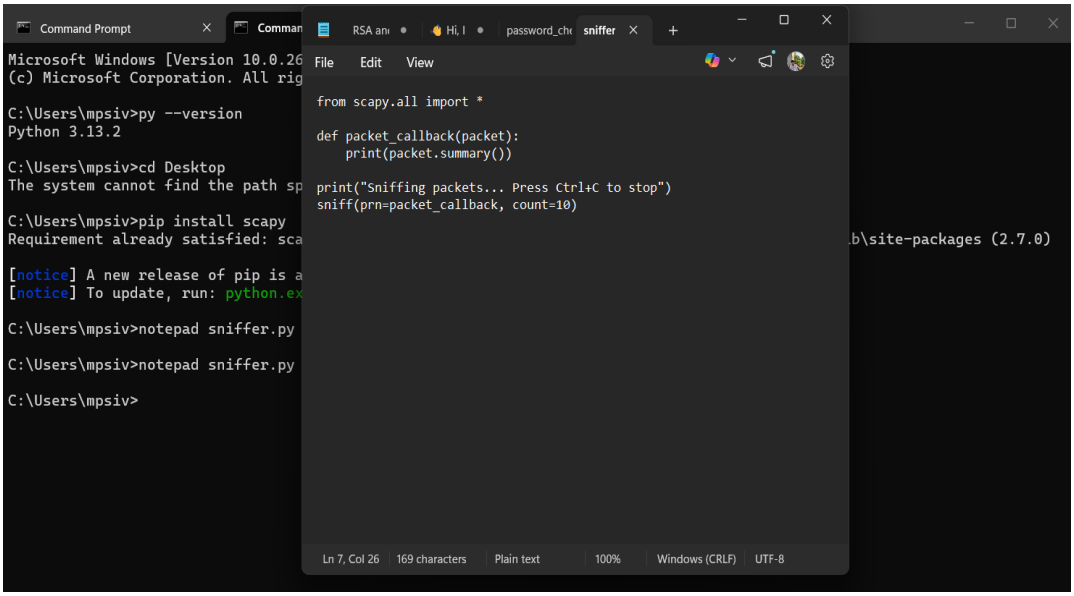
```
Microsoft Windows [Version 10.0.26200.8039]
(c) Microsoft Corporation. All rights reserved.

C:\Users\mpsiv>py --version
Python 3.13.2

C:\Users\mpsiv>cd Desktop
The system cannot find the path specified.
```

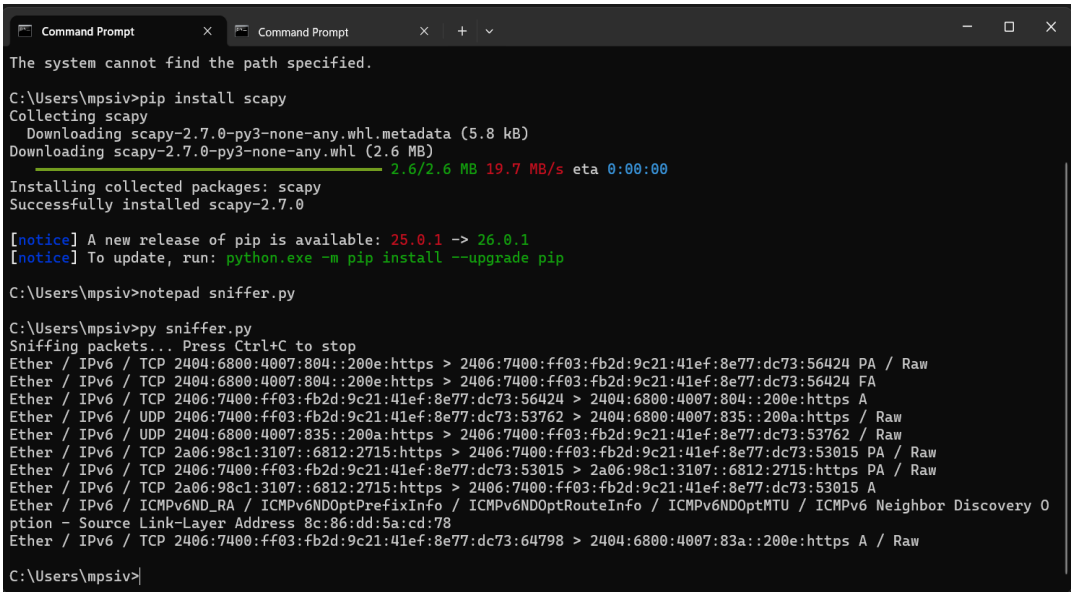
7. Step 4 – Write Code

The following code is used to sniff packets.



8. Step 5 – Run Program

Command: py sniffer.py



9. Output

The program captures network packets and displays protocol details such as TCP, UDP, and IPv6.

10. Verification

The successful output confirms that the Packet Sniffer & Analyzer is working correctly.